NEET UG STANDARD | BIOLOGY — CELL STRUCTURE & FUNCTION

Biomolecules: Chemical Composition, Structure, and Enzymes

Comprehensive Self-Contained Study Module, Structural Illustrations, Worked Examples, and Original NEET UG Practice Assessment

1. Learning Objectives & Chapter Overview

Despite the vast diversity of living organisms in our biosphere, their underlying chemical composition and metabolic pathways show remarkable unity. Whether we analyse a plant tissue, an animal tissue, or a microbial paste, we encounter the same fundamental elements and classes of organic and inorganic compounds. This tutorial systematically builds your conceptual and factual command over cellular chemistry for the **NEET UG** examination.

Key Learning Objectives

- Compare the elemental composition of living matter with non-living matter (Earth's crust) and understand wet weight, dry weight, and ash analysis.
- Distinguish between the trichloroacetic acid ($\mathrm{Cl_3CCOOH}$) soluble filtrate (micromolecules) and acid-insoluble retentate (biomacromolecules), and explain why lipids appear in the acid-insoluble fraction despite their low molecular weight.
- Recognize the chemical structures and classifications of amino acids (including zwitterionic forms), lipids (fatty acids, glycerol, glycerides, phospholipids, cholesterol), sugars, nitrogenous bases, nucleosides, and nucleotides.
- Differentiate primary metabolites from secondary metabolites and memorize key examples of secondary metabolites by category.
- Master the structural organization of proteins (primary, secondary, tertiary, quaternary), polysaccharides (cellulose, starch, glycogen, inulin, chitin), and nucleic acids (DNA and RNA).
- Understand enzyme catalysis, activation energy profiles, the catalytic cycle, factors affecting enzyme activity (T, pH, substrate concentration, competitive inhibition), the 6-class enzyme classification system, and co-factors (prosthetic groups, co-enzymes, metal ions).

2. Core Concepts and Detailed Explanations

2.1 Elemental Composition: Living Tissue vs. Non-Living Matter

When elemental analysis is performed on living tissue (such as a plant leaf, animal liver, or microbial paste) and compared against a sample of non-living matter (such as the Earth's crust), we obtain lists of elements like carbon, hydrogen, oxygen, nitrogen, sulphur, sodium, calcium, magnesium, and silicon.

- **Qualitative Similarity:** In absolute terms, there is no difference in the types of elements found; all elements present in a sample of the Earth's crust are also present in a sample of living tissue.
- **Quantitative Difference:** A closer examination reveals that the **relative abundance of carbon, hydrogen, and oxygen** with respect to other elements is significantly higher in any living organism than in the Earth's crust.

Tutorial Table 1: Comparison of Elements Present in Non-Living and Living Matter

Element	% Weight of Earth's Crust	% Weight of Human Body
Hydrogen (H)	0.14	9.5
Carbon (C)	0.03	18.5
Oxygen (O)	46.6	65.0
Nitrogen (N)	very little	3.3
Sulphur (S)	0.03	0.3
Sodium (Na)	2.8	0.2
Calcium (Ca)	3.6	1.5
Magnesium (Mg)	2.1	0.1
Silicon (Si)	27.7	negligible

2.2 How to Analyse Chemical Composition?

A. Analysis of Organic Constituents (Trichloroacetic Acid Fractionation)

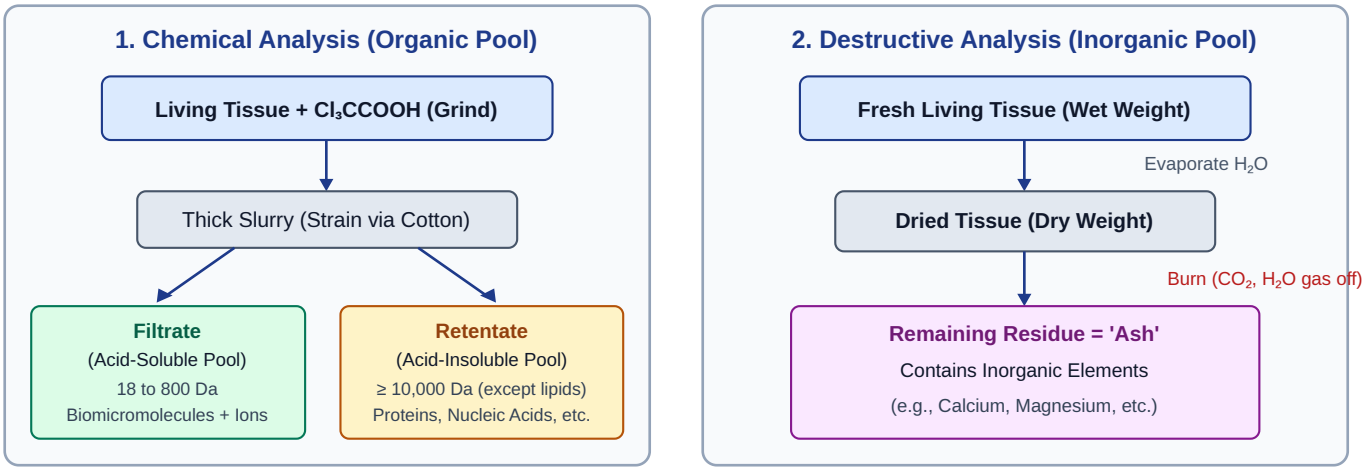
All carbon compounds obtained from living tissues are called **biomolecules**. To separate and study them:

1. Take any living tissue (e.g., a vegetable or a piece of liver) and grind it in **trichloroacetic acid (**$\mathrm{Cl_3CCOOH}$**)** using a mortar and a pestle to obtain a thick slurry.
2. Strain the slurry through cheesecloth or cotton to obtain two distinct fractions:
   - **The Filtrate (Acid-Soluble Pool):** Contains thousands of small organic compounds (molecular weight ranging from 18 to around $800\ \mathrm{Da}$) as well as inorganic ions/compounds like sulphate ($\mathrm{SO_4^{2-}}$) and phosphate ($\mathrm{PO_4^{3-}}$). It roughly represents the cytoplasmic composition.
   - **The Retentate (Acid-Insoluble Fraction):** Contains macromolecules from the cytoplasm and cellular organelles—specifically proteins, nucleic acids, polysaccharides, and lipids (in the form of disrupted membrane vesicles).
3. Extracts are then subjected to separation techniques to isolate and purify individual compounds. Analytical techniques determine the molecular formula and probable structure of each purified compound.

B. Analysis of Inorganic Constituents (Destructive Ash Analysis)

To identify inorganic elements and compounds in living organisms, a destructive experiment is performed:

1. **Wet Weight:** Weigh a small amount of fresh living tissue (such as a leaf or liver).
2. **Dry Weight:** Dry the tissue completely so that all water evaporates. The remaining material gives the dry weight.
3. **Ash:** Burn the dried tissue completely. All carbon compounds are oxidised to gaseous forms ($\mathrm{CO_2}$ and water vapour) and are removed. The remaining residue is called **ash**. This ash contains inorganic elements such as calcium (Ca), magnesium (Mg), etc.



Tutorial Diagram 1: Protocols for analysing organic fractions (left) and inorganic constituents (right) in living tissues.

Tutorial Table 2: Representative Inorganic Constituents of Living Tissues

Component	Formula / Representation
Sodium	Na^{+}
Potassium	K^{+}
Calcium	Ca^{++}
Magnesium	Mg^{++}
Water	$\mathrm{H_2O}$
Compounds	$\mathrm{NaCl}, \mathrm{CaCO_3}, \mathrm{PO_4^{3-}}, \mathrm{SO_4^{2-}}$

2.3 Small Molecular Weight Organic Compounds (Biomicromolecules)

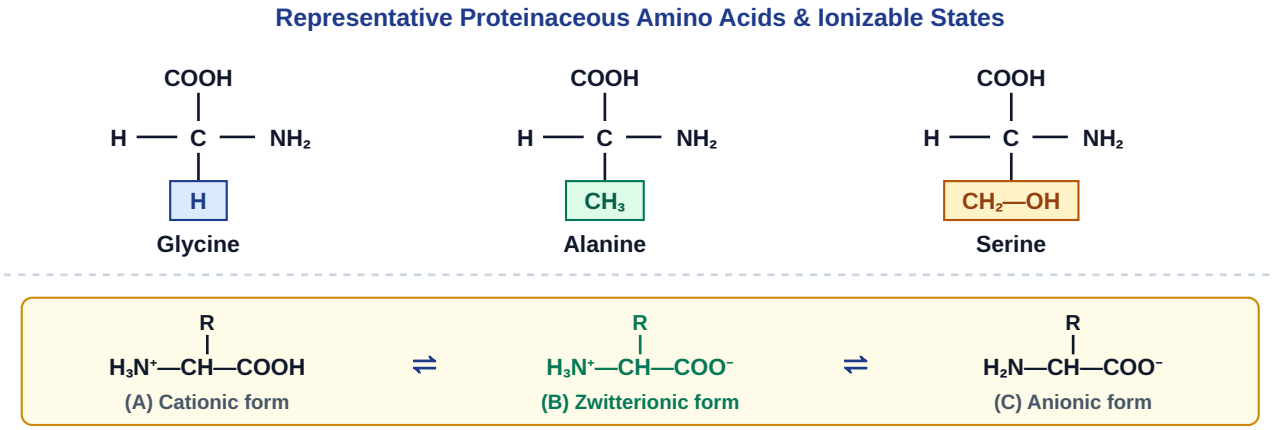
From a chemical perspective, one identifies functional groups such as aldehydes, ketones, and aromatic rings. From a biological perspective, small organic biomolecules are classified into **amino acids**, **fatty acids/lipids**, **sugars**, and **nucleotide bases/nucleosides/nucleotides**.

A. Amino Acids

- **Structure:** Amino acids are organic compounds containing an amino group ($-\mathrm{NH_2}$) and an acidic carboxyl group ($-\mathrm{COOH}$) as substituents on the *same* carbon atom—the α**-carbon**. Hence, they are called α**-amino acids**.
- **Substituted Methanes:** They are substituted methanes with four substituent groups occupying the four valency positions of the α-carbon:
  1. Hydrogen ($-\mathrm{H}$)
  2. Carboxyl group ($-\mathrm{COOH}$)
  3. Amino group ($-\mathrm{NH_2}$)
  4. Variable group designated as the R **group**
- **Proteinaceous Amino Acids:** While many amino acids exist in nature based on the R group, those occurring in proteins are of only **20 types**:
  - If $\mathrm{R} = -\mathrm{H}$, the amino acid is **Glycine** (simplest amino acid).
  - If $\mathrm{R} = -\mathrm{CH_3}$ (methyl group), the amino acid is **Alanine**.
  - If $\mathrm{R} = -\mathrm{CH_2} - \mathrm{OH}$ (hydroxy methyl group), the amino acid is **Serine**.
- **Classification by Functional Groups:** The chemical and physical properties of amino acids depend on the amino, carboxyl, and R functional groups:

- **Acidic amino acids** (extra carboxyl group): e.g., **Glutamic acid**.
- **Basic amino acids** (extra amino group): e.g., **Lysine**.
- **Neutral amino acids** (equal amino and carboxyl groups): e.g., **Valine**.
- **Aromatic amino acids** (possess aromatic ring): e.g., **Tyrosine, Phenylalanine, Tryptophan**.

- **Ionizable Nature & Zwitterionic Form:** Because the $-\mathrm{NH_2}$ and $-\mathrm{COOH}$ groups are ionizable, the structure of amino acids changes in solutions of different pH. The dipolar ion carrying both a positive charge ($-\mathrm{NH_3^+}$) and a negative charge ($-\mathrm{COO^-}$) is called the **zwitterionic form**.



Tutorial Diagram 2: Structures of Glycine, Alanine, and Serine (top), and pH-dependent ionization of amino acids showing the Zwitterionic form (B) (bottom).

B. Lipids

Lipids are generally **water-insoluble** compounds. They include:

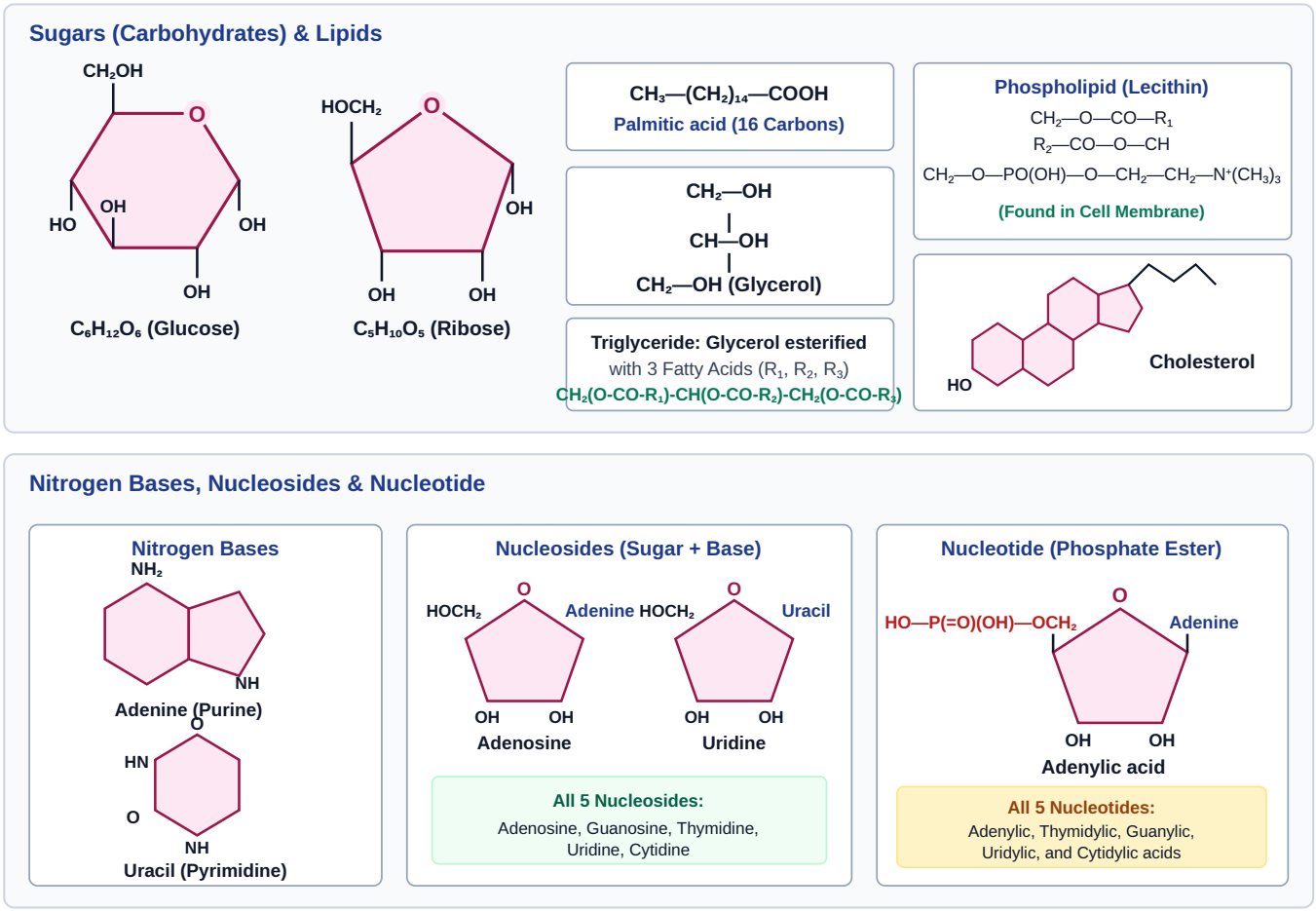
- **Simple Fatty Acids:** A fatty acid has a carboxyl group ($-\mathrm{COOH}$) attached to an R group. The R group can be a methyl ($-\mathrm{CH_3}$), an ethyl ($-\mathrm{C_2H_5}$), or a higher number of $-\mathrm{CH_2}$ groups (1 carbon to 19 carbons).
  - **Palmitic acid:** Has **16 carbons** including the carboxyl carbon ($\mathrm{CH_3 - (CH_2)_{14} - COOH}$).
  - **Arachidonic acid:** Has **20 carbon atoms** including the carboxyl carbon.
  - Fatty acids can be **saturated** (without double bonds) or **unsaturated** (with one or more $\mathrm{C = C}$ double bonds).
- **Glycerol:** A simple lipid chemically known as **trihydroxy propane** ($\mathrm{CH_2OH - CHOH - CH_2OH}$).
- **Glycerides (Fats and Oils):** Many lipids have both glycerol and fatty acids, where fatty acids are **esterified** with glycerol. Depending on the number of fatty acids attached (1, 2, or 3), they are called **monoglycerides**, **diglycerides**, or **triglycerides**.
  - Based on melting point, glycerides are called fats or oils. **Oils have a lower melting point** (e.g., **gingelly oil**) and hence remain as liquid oil in winters, whereas fats solidify at room/winter temperatures (e.g., butter, ghee).
- **Phospholipids:** Lipids containing phosphorus and a phosphorylated organic compound. They are found in the **cell membrane**. **Lecithin** is a prime example.
- **Complex & Steroid Lipids:** Neural tissues contain lipids with more complex structures. **Cholesterol** is another important lipid with a four-ring fused steroid nucleus and a hydroxyl ($-\mathrm{OH}$) group.

C. Sugars, Nitrogenous Bases, Nucleosides, and Nucleotides

- **Monosaccharide Sugars:** Key examples include **Glucose** ($\mathrm{C_6H_{12}O_6}$, a hexose forming a 6-membered pyranose ring) and **Ribose** ($\mathrm{C_5H_{10}O_5}$, a pentose forming a 5-membered furanose ring) or **2'-deoxyribose**.
- **Nitrogenous Bases:** Heterocyclic carbon-nitrogen ring compounds:

- **Substituted Purines** (dicyclic/two-ring structure): **Adenine (A)** and **Guanine (G)**.
- **Substituted Pyrimidines** (monocyclic/single-ring structure): **Cytosine (C)**, **Uracil (U)**, and **Thymine (T)**.

- **Nucleosides (Base + Sugar):** When a nitrogenous base is found attached to a sugar, it is called a **nucleoside**. Examples: **Adenosine, Guanosine, Thymidine, Uridine, and Cytidine**.
- **Nucleotides (Base + Sugar + Phosphate):** If a phosphate group (phosphoric acid) is also found esterified to the sugar of a nucleoside, it is called a **nucleotide**. Examples: **Adenylic acid, Thymidylic acid, Guanylic acid, Uridylic acid, and Cytidylic acid**. Nucleic acids like DNA and RNA consist solely of nucleotides.



Tutorial Diagram 3: Diagrammatic representation of small molecular weight organic compounds in living tissues (sugars, lipids, nitrogen bases, nucleosides, and nucleotides).

2.4 Primary and Secondary Metabolites

All biomolecules involved in cellular reactions can be called **metabolites**.

- **Primary Metabolites:** Organic compounds such as amino acids, sugars, fatty acids, glycerol, nitrogen bases, nucleosides, and nucleotides that are present in animal, plant, and microbial tissues. They have **identifiable functions** and play **known roles in normal physiological processes**.
- **Secondary Metabolites:** When plant, fungal, and microbial cells are analysed, thousands of compounds other than primary metabolites are found—such as alkaloids, flavonoids, rubber, essential oils, antibiotics, coloured pigments, scents, gums, and spices.
  - We do not at the moment understand the exact role or functions of all secondary metabolites in the **host organisms**.
  - However, many of them are useful to **human welfare** (e.g., rubber, drugs, spices, scents, and pigments), and some have **ecological importance**.

Tutorial Table 3: Categories and Examples of Secondary Metabolites (High-Yield NEET Table)

Category	Examples
Pigments	Carotenoids, Anthocyanins, etc.
Alkaloids	Morphine, Codeine, etc.
Terpenoides	Monoterpenes, Diterpenes, etc.
Essential oils	Lemon grass oil, etc.
Toxins	Abrin, Ricin
Lectins	Concanavalin A
Drugs	Vinblastin, curcumin, etc.
Polymeric substances	Rubber, gums, cellulose

2.5 Biomacromolecules and the Lipid Paradox

Biomolecules in living organisms are divided into two groups based on molecular weight and solubility in trichloroacetic acid:

1. **Micromolecules (or simply Biomolecules):** Found in the **acid-soluble pool**; molecular weights range from 18 **to around** $800\ \mathrm{Da}$ (less than $1000\ \mathrm{Da}$).
2. **Macromolecules (Biomacromolecules):** Found in the **acid-insoluble fraction**. This fraction has only four types of organic compounds: **proteins, nucleic acids, polysaccharides, and lipids**. With the exception of lipids, these compounds are polymeric substances with molecular weights in the range of $10,000\ \mathrm{Da}$ **and above**.

> **Why Do Lipids Come Under the Acid-Insoluble (Macromolecular) Fraction?**
>
> Lipids have molecular weights that **do not exceed** $800\ \mathrm{Da}$ and are not polymeric substances. Therefore, **lipids are not strictly macromolecules**.
>
> However, lipids are present not only as individual molecules but are also arranged into structures like the **cell membrane and other organelle membranes**. When we grind a tissue, we disrupt the cell structure. Cell membranes and other membranes are broken into pieces and form **vesicles which are not water-soluble**. Consequently, these water-insoluble membrane fragments in the form of vesicles get separated along with the acid-insoluble pool (retentate) and hence appear in the macromolecular fraction.

When we arrange the chemical composition of living tissue in order of abundance, **water is the most abundant chemical in living organisms** (70–90% of total cellular mass), followed by proteins (10–15%), nucleic acids (5–7%), carbohydrates (3%), lipids (2%), and ions (1%).

Tutorial Table 4: Average Composition of Cells

Component	% of the Total Cellular Mass
Water	70–90
Proteins	10–15
Nucleic acids	5–7
Carbohydrates	3
Lipids	2
Ions	1

2.6 Proteins and Their Levels of Structure

Proteins are polypeptides—linear chains of amino acids linked by **peptide bonds**. Because there are 20 different types of amino acids (e.g., alanine, cysteine, proline, tryptophan, lysine, etc.) that build proteins, **a**

protein is a heteropolymer and not a homopolymer (a homopolymer has only one type of monomer repeating n times).

- **Essential vs. Non-Essential Amino Acids:**
  - **Essential amino acids:** Essential for health and cannot be synthesized by our body; they must be supplied through our **diet/food** (dietary proteins).
  - **Non-essential amino acids:** Those which our body can synthesize on its own.
- **Most Abundant Proteins:**
  - **Collagen** is the most abundant protein in the **animal world**.
  - **Ribulose bisphosphate Carboxylase-Oxygenase (RuBisCO)** is the most abundant protein in the **whole of the biosphere**.

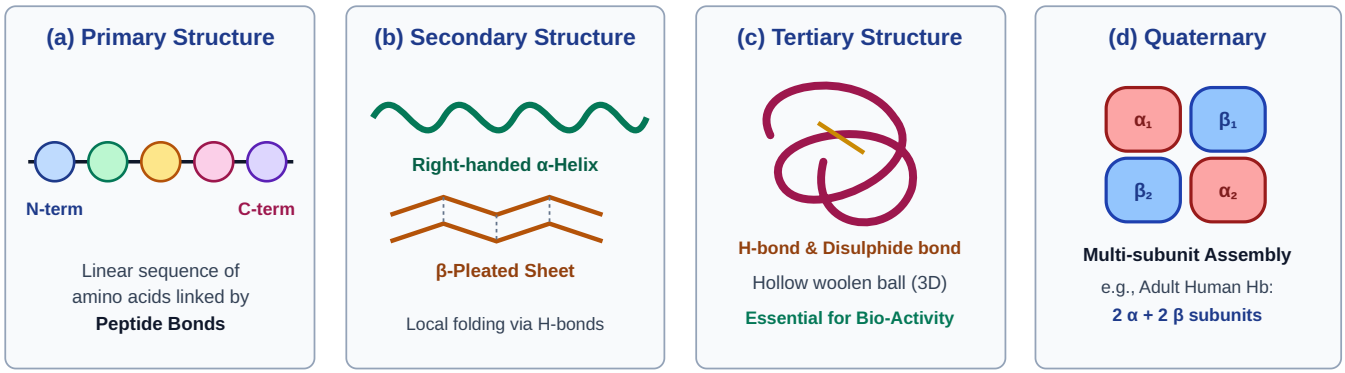
Tutorial Table 5: Important Proteins and Their Biological Functions

Protein	Functions
Collagen	Intercellular ground substance
Trypsin	Enzyme
Insulin	Hormone
Antibody	Fights infectious agents
Receptor	Sensory reception (smell, taste, hormone, etc.)
GLUT-4	Enables glucose transport into cells

Four Levels of Protein Structure

While inorganic chemists express structure as molecular formulae (e.g., NaCl, $\mathrm{MgCl_2}$), organic chemists write 2-dimensional views (e.g., benzene, naphthalene), and physicists focus on 3-dimensional views, biologists describe protein structure at **four levels**:

1. **Primary Structure:** The sequence of amino acids, i.e., the **positional information** in a protein (which is the first amino acid, which is second, and so on). Imagined as a line:
   - The **left end** is represented by the first amino acid, called the **N-terminal amino acid**.
   - The **right end** is represented by the last amino acid, called the **C-terminal amino acid**.
2. **Secondary Structure:** A protein thread does not exist throughout as an extended rigid rod. Portions of the thread are folded in the form of a **helix** (similar to a revolving staircase) or a β**-pleated sheet**. In proteins, **only right-handed helices** are observed.
3. **Tertiary Structure:** The long protein chain is also folded upon itself like a **hollow woolen ball**, giving rise to a 3-dimensional view stabilized by interactions such as hydrogen bonds and disulphide bonds. **Tertiary structure is absolutely necessary for the many biological activities of proteins** (such as enzymatic activity).
4. **Quaternary Structure:** Some proteins are an assembly of more than one polypeptide or subunit. The manner in which these individual folded polypeptides or subunits are arranged with respect to each other (e.g., linear string of spheres, spheres arranged one upon each other in the form of a cube or plate) is the architecture or **quaternary structure**.
   - *Example:* Adult human haemoglobin (Hb) consists of **4 subunits**—two identical subunits of α **type** and two identical subunits of β **type** ($\alpha_2\beta_2$).

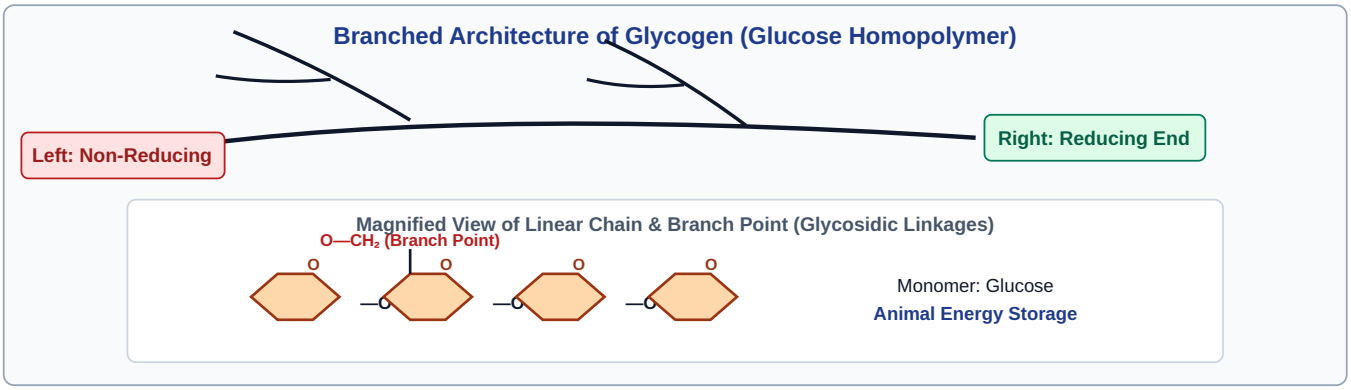


Tutorial Diagram 4: The four hierarchical levels of protein structure—Primary, Secondary, Tertiary, and Quaternary.

2.7 Polysaccharides (Carbohydrates)

Polysaccharides are another class of macromolecules in the acid-insoluble pellet. They are long chains of sugars—threads containing monosaccharides as building blocks linked by glycosidic bonds.

- **Cellulose:** A polymeric polysaccharide consisting of only one type of monosaccharide—**glucose**. Hence, cellulose is a **homopolymer**. Plant cell walls are made of cellulose; paper made from plant pulp and cotton fibre is cellulosic. Cellulose **does not contain complex helices** and hence **cannot hold** I_2 molecules.
- **Starch:** A homopolymer of glucose present as a storehouse of energy in **plant tissues**. Starch forms **helical secondary structures** and can hold I_2 (iodine) molecules in the helical portion, giving a characteristic **blue colour** (starch-I_2 complex).
- **Glycogen:** The animal variant of glucose homopolymer, serving as the energy storehouse in animal tissues. It is highly branched. In a polysaccharide chain like glycogen, the **right end is called the reducing end** and the **left end is called the non-reducing end**.
- **Inulin:** A homopolymer of **fructose**.
- **Complex Polysaccharides (Chitin):** In nature, complex polysaccharides have amino-sugars and chemically modified sugars as building blocks (e.g., **glucosamine**, **N-acetyl galactosamine**). The exoskeletons of arthropods have a complex polysaccharide called **chitin**. Complex polysaccharides are **mostly homopolymers**.



Tutorial Diagram 5: Diagrammatic representation of a portion of glycogen showing branching, glycosidic linkages, non-reducing left end, and reducing right end.

2.8 Nucleic Acids

Nucleic acids are **polynucleotides** found in the acid-insoluble fraction. Together with polysaccharides and polypeptides, they comprise the **true macromolecular fraction** of any living tissue or cell.

- **Building Block (Nucleotide):** Every nucleotide has three chemically distinct components:
  1. A **heterocyclic compound** (nitrogenous base):

- **Substituted Purines:** Adenine and Guanine.
- **Substituted Pyrimidines:** Uracil, Cytosine, and Thymine.

2. A **monosaccharide pentose sugar**: Either **ribose** or **2'-deoxyribose**.
3. A **phosphoric acid or phosphate** group.

- **DNA vs. RNA:** A nucleic acid containing deoxyribose is called **deoxyribonucleic acid (DNA)**, while that which contains ribose is called **ribonucleic acid (RNA)**. Both function as genetic material, carrying hereditary information from parental generation to progeny.

2.9 Enzymes: Biological Catalysts

- **Chemical Nature:** Almost all enzymes are **proteins**. However, there are some **nucleic acids that behave like enzymes**; these are called **ribozymes**.
- **Active Site:** Like any protein, an enzyme has primary, secondary, and tertiary structures. In the tertiary structure, the polypeptide chain folds upon itself and criss-crosses, creating many crevices or pockets. One such pocket is the **active site**—a crevice or pocket into which the **substrate** fits.
- **Comparison with Inorganic Catalysts:** Inorganic catalysts work efficiently at high temperatures and high pressures, whereas most enzymes get damaged (denatured) at high temperatures (**above** $40\ ^\circ\mathrm{C}$).
  - *Exception (Thermophiles):* Enzymes isolated from organisms living under extremely high temperatures (e.g., hot vents and sulphur springs) are thermally stable and retain their catalytic power even at high temperatures (**up to** $80\ ^\circ\mathrm{C}$–$90\ ^\circ\mathrm{C}$).

A. Physical Changes vs. Chemical Reactions & Reaction Rates

- **Physical Process:** A change in shape without breaking bonds, or a change in state of matter (e.g., ice melting into water, or water becoming vapour).
- **Chemical Reaction:** A transformation in which old chemical bonds are broken and new bonds are formed.
  - *Inorganic example:* $\mathrm{Ba(OH)_2 + H_2SO_4 \rightarrow BaSO_4 + 2H_2O}$
  - *Organic example:* Hydrolysis of starch into glucose.

- **Rate (or Velocity) of a Process:** Refers to the amount of product formed per unit time:

$$\text{rate} = \frac{\delta P}{\delta t}$$

  Rate is called **velocity** if the direction is specified.
- **Temperature Rule of Thumb:** Rates of physical and chemical processes are influenced by temperature. A general rule of thumb is that the **rate doubles or decreases by half for every** $10\ ^\circ\mathrm{C}$ **change in either direction**.
- **Dramatic Acceleration by Carbonic Anhydrase:** Consider the hydration of carbon dioxide:

$$\mathrm{CO_2 + H_2O \xrightarrow{\text{Carbonic anhydrase}} H_2CO_3}$$

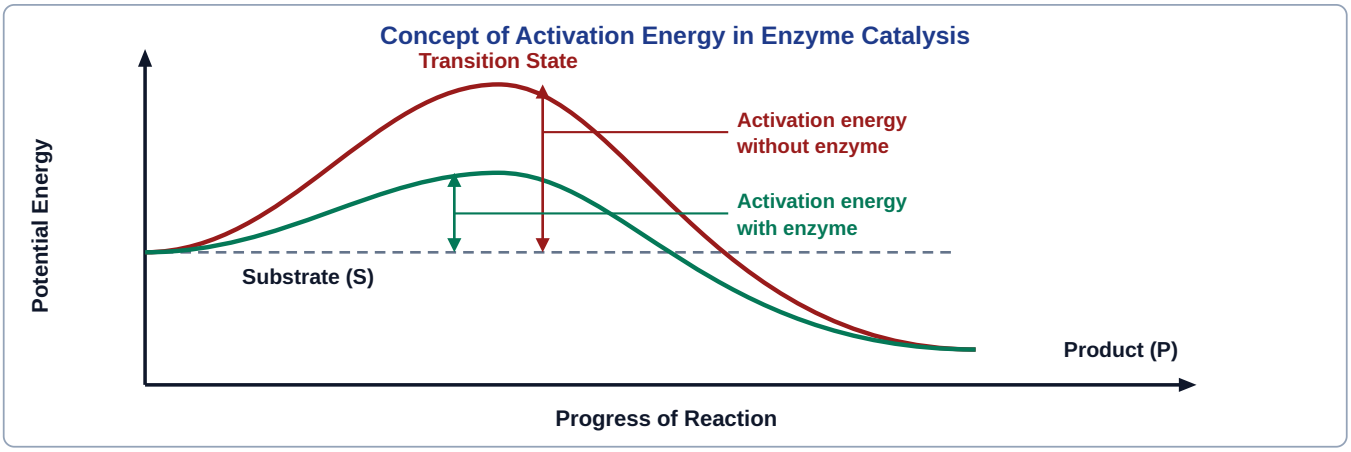
  - **Uncatalysed (absence of enzyme):** Very slow—about 200 **molecules of** $\mathrm{H_2CO_3}$ **formed per hour**.
  - **Catalysed (with cytoplasmic carbonic anhydrase):** About $600,000$ **molecules formed every second**.
  - **Acceleration Factor:** The enzyme accelerates the reaction rate by about $10\ \text{million}$ **times** (10^7 times)!

- **Metabolic Pathways:** A multistep chemical reaction where each step is catalysed by the same enzyme complex or different enzymes is called a **metabolic pathway**. For example, the conversion of glucose to pyruvic acid ($\mathrm{C_6H_{12}O_6 + O_2 \rightarrow 2C_3H_4O_3 + 2H_2O}$) occurs through **10 different enzyme-catalysed metabolic reactions** (glycolysis). Depending on conditions, one or two additional reactions yield different end products:
  - **Normal aerobic conditions:** Pyruvic acid is formed.
  - **Human skeletal muscle (anaerobic conditions):** Lactic acid is formed.
  - **Yeast (fermentation):** Ethanol (alcohol) is produced.

B. Activation Energy and Transition State

In an enzyme-catalysed reaction ($\mathrm{S} \rightarrow \mathrm{P}$), the substrate ($\mathrm{S}$) diffuses toward and binds to the active site of the enzyme (E), forming an obligatory, transient ES **complex**. While bound, a new, unstable high-energy structure of the substrate called the **transition state structure** is formed before bond breaking/making completes and product (P) is released.

- If the potential energy level of P is lower than that of S, the reaction is **exothermic** (spontaneous; no external heating needed to form product). If P is higher than S, it is **endothermic** (energy-requiring).
- In both exothermic and endothermic reactions, S must pass through the higher-energy transition state. The difference in average energy content of S from that of this transition state is called **activation energy**.
- **Enzymes lower this activation energy barrier**, making the transition from S to P much easier and faster.



Tutorial Diagram 6: Potential energy profile showing how an enzyme lowers the activation energy barrier for an exothermic reaction (P at lower energy than S).

C. Nature of Enzyme Action (The 4-Step Catalytic Cycle)

The reaction sequence is represented as:

$$\mathrm{E} + \mathrm{S} \rightleftharpoons \mathrm{ES} \longrightarrow \mathrm{EP} \longrightarrow \mathrm{E} + \mathrm{P}$$

1. First, the substrate (S) binds to the active site of the enzyme (E), fitting into the active site.
2. The binding of the substrate induces the enzyme to **alter its shape**, fitting more tightly around the substrate.
3. The active site of the enzyme, now in close proximity to the substrate, **breaks the chemical bonds** of the substrate and the new **enzyme-product (EP) complex** is formed.
4. The enzyme **releases the products (P)** of the reaction, and the free enzyme (E) is ready to bind to another molecule of the substrate and run through the catalytic cycle once again.

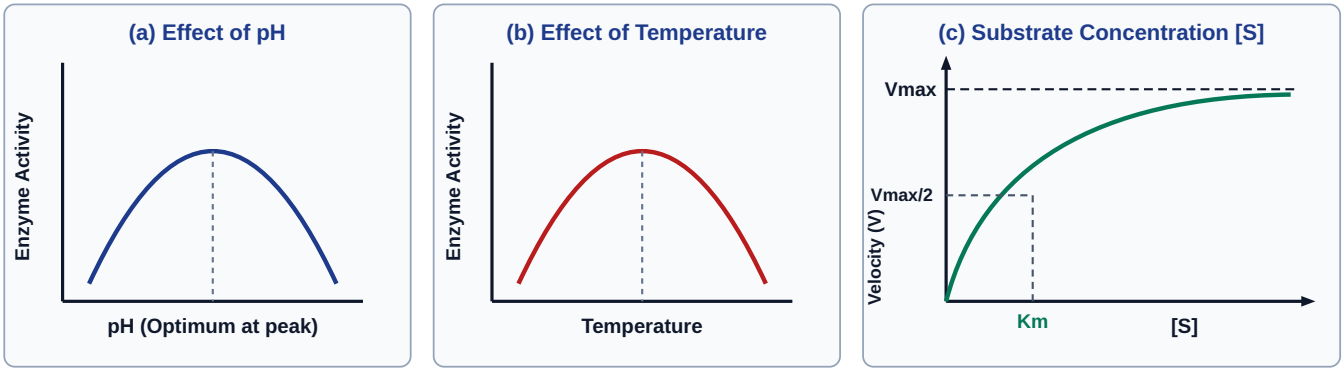
D. Factors Affecting Enzyme Activity

- **Temperature and pH:** Enzymes function in a narrow range of temperature and pH, showing highest activity at a specific **optimum temperature** and **optimum pH** (bell-shaped curves). Activity declines both below and above the optimum value:
  - **Low temperature** preserves the enzyme in a **temporarily inactive state**.
  - **High temperature** destroys enzymatic activity because proteins are **denatured by heat** (disrupting tertiary structure).

- **Concentration of Substrate ($[\mathrm{S}]$):** With an increase in substrate concentration, the velocity of the enzymatic reaction rises at first. The reaction ultimately reaches a **maximum velocity ($V_{\max}$)** which is not exceeded by any further rise in substrate concentration.
  - *Reason:* Enzyme molecules are fewer than substrate molecules; after **saturation** of all enzyme molecules, there are **no free enzyme molecules** left to bind with additional substrate molecules.

- **Michaelis Constant (K_m):** The substrate concentration at which the chemical reaction attains **half its maximum velocity ($\frac{V_{\max}}{2}$)**.

- **Inhibition and Competitive Inhibitors:** When the binding of a specific chemical shuts off enzyme activity, the process is called **inhibition** and the chemical is an **inhibitor**.
  - **Competitive Inhibitor:** When the inhibitor closely resembles the substrate in its molecular structure, it competes with the substrate for the **substrate-binding (active) site** of the enzyme. Consequently, the substrate cannot bind and enzyme action declines.
  - *Classic Example:* Inhibition of **succinic dehydrogenase** by **malonate**, which closely resembles the substrate **succinate** in structure.
  - *Application:* Competitive inhibitors are often used in the **control of bacterial pathogens**.



Tutorial Diagram 7: Effect of change in (a) pH, (b) Temperature, and (c) Substrate concentration [S] on enzyme activity.

E. Classification and Nomenclature of Enzymes

Enzymes are divided into **6 classes** based on the type of reactions they catalyse, each with **4–13 subclasses**, and named accordingly by a **four-digit number**:

1. **Class 1 — Oxidoreductases / Dehydrogenases:** Catalyse oxidoreduction between two substrates S and S':
$$\mathrm{S\ reduced} + \mathrm{S}'\ \mathrm{oxidised} \longrightarrow \mathrm{S\ oxidised} + \mathrm{S}'\ \mathrm{reduced}$$
2. **Class 2 — Transferases:** Catalyse transfer of a group G (**other than hydrogen**) between a pair of substrates S and S':
$$\mathrm{S-G} + \mathrm{S}' \longrightarrow \mathrm{S} + \mathrm{S}' - \mathrm{G}$$
3. **Class 3 — Hydrolases:** Catalyse hydrolysis of ester, ether, peptide, glycosidic, $\mathrm{C-C}$, C-halide, or $\mathrm{P-N}$ bonds.
4. **Class 4 — Lyases:** Catalyse removal of groups from substrates by mechanisms **other than hydrolysis**, leaving double bonds:
$$\begin{array}{cc} \mathrm{X} & \mathrm{Y} \\ | & | \\ \mathrm{C} & \mathrm{C} \end{array} \longrightarrow \mathrm{X-Y} + \mathrm{C=C}$$
5. **Class 5 — Isomerases:** Include all enzymes catalysing inter-conversion of optical, geometric, or positional isomers.
6. **Class 6 — Ligases:** Catalyse the linking together of 2 compounds (e.g., enzymes which catalyse joining of $\mathrm{C-O}$, $\mathrm{C-S}$, $\mathrm{C-N}$, $\mathrm{P-O}$, etc. bonds).

F. Co-Factors

In many enzymes, non-protein constituents called **co-factors** are bound to the enzyme to make it catalytically active. In these instances, the **protein portion of the enzyme is called the apoenzyme** (Holoenzyme = Apoenzyme + Co-factor). **Catalytic activity is lost when the co-factor is removed** from the enzyme.

Three kinds of co-factors are identified:

Tutorial Table 6: Comparison of the Three Types of Enzyme Co-Factors

Co-Factor Type	Chemical Nature & Binding	Key Examples & Details
1. Prosthetic Groups	**Organic compounds** that are **tightly bound** to the apoenzyme.	In **peroxidase** and **catalase** (which catalyse the breakdown of hydrogen peroxide H_2O_2 to water and oxygen), **haem** is the prosthetic group and is part of the active site.
2. Co-enzymes	**Organic compounds** whose association with the apoenzyme is **only transient**, usually occurring during the course of catalysis. Serve as co-factors in many different enzyme-catalysed reactions.	The essential chemical components of many co-enzymes are **vitamins**. E.g., co-enzymes **NAD** (nicotinamide adenine dinucleotide) and **NADP** contain the vitamin **niacin**.
3. Metal Ions	Inorganic metal cations that form **coordination bonds with side chains at the active site** and at the same time form **one or more coordination bonds with the substrate**.	**Zinc** (Zn^{2+}) is a co-factor for the proteolytic enzyme **carboxypeptidase**.

3. Key Definitions, Equations, and High-Yield Results

- **Rate of Reaction:** $\text{rate} = \dfrac{\delta P}{\delta t}$ (called *velocity* when direction is specified).
- **Temperature Rule of Thumb (Q_{10} concept):** Rate doubles or decreases by half for every $10\,^{\circ}\text{C}$ change in either direction.
- **Carbonic Anhydrase Acceleration:**

  $$CO_2 + H_2O \xrightarrow{\text{Carbonic anhydrase}} H_2CO_3$$

  Uncatalysed rate $= 200$ molecules/hour; Catalysed rate $= 600,000$ molecules/second ($\text{Acceleration} = 10$ million times).
- **Enzyme-Substrate Catalytic Sequence:**

  $$E + S \rightleftharpoons ES \longrightarrow EP \longrightarrow E + P$$
- **Glycolysis (10-Step Metabolic Pathway):**

  $$C_6H_{12}O_6 + O_2 \longrightarrow 2C_3H_4O_3\ (\text{Pyruvic acid}) + 2H_2O$$
- **Michaelis Constant (K_m):** Substrate concentration $[S]$ at which reaction velocity $V = \dfrac{V_{\max}}{2}$.

4. Conceptual Connections & Comparative Matrices

Tutorial Table 7: Master Comparison of Major Polysaccharides

Polysaccharide	Monomer / Building Block	Polymer Type	Key Structural & Functional Features
Cellulose	Glucose	Homopolymer	Unbranched; forms plant cell walls, cotton fibre, paper; no complex helices; does **not** give blue colour with I_2.
Starch	Glucose	Homopolymer	Plant energy storage; forms **helical secondary structures**; holds I_2 molecules to give a **blue colour**.
Glycogen	Glucose	Homopolymer	Animal energy storage; branched chain; **right end = reducing end**, **left end = non-reducing end**.
Inulin	Fructose	Homopolymer	Polymer of fructose units.
Chitin	Chemically modified sugar / amino-sugar (N-acetyl glucosamine)	Homopolymer (complex polysaccharide)	Forms the **exoskeletons of arthropods**.

5. Worked Examples (Step-by-Step Application)

Worked Example 1: Quantitative Comparison of Catalysed vs. Uncatalysed Reaction Rates

Problem: In the absence of any enzyme, 200 molecules of carbonic acid ($\mathrm{H_2CO_3}$) are formed per hour from $\mathrm{CO_2}$ and $\mathrm{H_2O}$. In the presence of cytoplasmic carbonic anhydrase, $600,000$ molecules of $\mathrm{H_2CO_3}$ are formed every second. Calculate the exact ratio of the catalysed rate to the uncatalysed rate when both are expressed in molecules formed per hour.

Solution:

1. **Uncatalysed rate per hour:**
$$R_{\text{uncatalysed}} = 200 \text{ molecules/hour}$$
2. **Convert catalysed rate from molecules/second to molecules/hour:** Since $1 \text{ hour} = 60 \text{ minutes} \times 60 \text{ seconds} = 3,600 \text{ seconds}$:
$$R_{\text{catalysed}} = 600,000 \text{ molecules/s} \times 3,600 \text{ s/hour} = 2.16 \times 10^9 \text{ molecules/hour}$$
3. **Ratio of catalysed to uncatalysed rate:**
$$\text{Ratio} = \frac{2.16 \times 10^9}{200} = 1.08 \times 10^7 \approx 10 \text{ million times}$$

Conclusion: Carbonic anhydrase accelerates the hydration of $\mathrm{CO_2}$ by approximately 10 million **times** (1.08×10^7).

Worked Example 2: Identifying Fractions in Trichloroacetic Acid ($\mathrm{Cl_3CCOOH}$) Extraction

Problem: A student grinds a piece of fresh liver tissue in trichloroacetic acid and strains the slurry through cotton. Predict whether each of the following components will be found predominantly in the *filtrate (acid-soluble pool)* or the *retentate (acid-insoluble fraction)*, and state why: (a) Alanine, (b) Lecithin, (c) RuBisCO/Collagen, (d) Adenylic acid, (e) Glycogen, (f) $\mathrm{SO_4^{2-}}$ and $\mathrm{PO_4^{3-}}$.

Solution:

- **(a) Alanine:** *Filtrate (acid-soluble pool)* — It is a monomeric amino acid with molecular weight well within 18–800 Da.
- **(b) Lecithin:** *Retentate (acid-insoluble fraction)* — Although lecithin is a phospholipid with molecular weight $< 800 \text{ Da}$, it is a component of cell membranes. Upon grinding, membranes break into water-insoluble vesicles that are retained in the acid-insoluble fraction.
- **(c) RuBisCO / Collagen:** *Retentate (acid-insoluble fraction)* — Polypeptide macromolecules with molecular weight $\geq 10,000 \text{ Da}$.
- **(d) Adenylic acid:** *Filtrate (acid-soluble pool)* — It is a single nucleotide monomer (molecular weight $< 1000 \text{ Da}$), unlike polymeric DNA or RNA.
- **(e) Glycogen:** *Retentate (acid-insoluble fraction)* — Polymeric polysaccharide macromolecule.
- **(f) $\mathrm{SO_4^{2-}}$ and $\mathrm{PO_4^{3-}}$:** *Filtrate (acid-soluble pool)* — Inorganic compounds/ions dissolve in the acid-soluble fraction.

Worked Example 3: Resolving Textbook Conceptual Exercises

Problem: Provide concise, exam-ready explanations for: (a) Composition of a triglyceride, (b) Why tertiary structure of proteins is vital, and (c) How gums differ from synthetic adhesives like Fevicol.

Solution:

- **(a) Composition of a Triglyceride:** A triglyceride is a glyceride lipid composed of **one molecule of glycerol** (trihydroxy propane) esterified with **three molecules of fatty acids** ($\mathrm{R_1COOH}$, $\mathrm{R_2COOH}$, $\mathrm{R_3COOH}$) via three ester bonds.
- **(b) Importance of Tertiary Structure:** In the tertiary structure, the long polypeptide chain folds upon itself like a hollow woolen ball (3D conformation), bringing distant amino acid side chains together to form crevices/pockets called **active sites**. This 3D tertiary structure is absolutely necessary for the biological and enzymatic activities of proteins.
- **(c) Natural Gums vs. Fevicol:** Natural gums are **polymeric secondary metabolites** (carbohydrates/polysaccharides) produced by plants, whereas Fevicol is a **synthetic resin adhesive** (polyvinyl acetate polymer) manufactured industrially from petrochemicals.

6. Original NEET UG Practice Questions with Worked Solutions

Note: All questions below are original practice questions designed strictly in accordance with the NEET UG examination pattern to test conceptual clarity, factual accuracy, and analytical reasoning.

Practice Question 1

Single-Correct MCQ | Elemental Analysis

When the percentage weight of elements in the human body is compared with that in the Earth's crust, which of the following sets of elements shows a higher percentage weight in the Earth's crust than in the human body?

(1) Carbon, Hydrogen, and Nitrogen

(2) Oxygen, Carbon, and Sulphur

(3) Silicon, Calcium, Sodium, and Magnesium

(4) Hydrogen, Oxygen, and Calcium

Correct Answer: (3)

Worked Solution: Comparing the elemental composition (% weight in Earth's crust vs. Human body): Silicon (27.7% vs. negligible), Calcium (3.6% vs. 1.5%), Sodium (2.8% vs. 0.2%), and Magnesium (2.1% vs. 0.1%) are all higher in the Earth's crust. In contrast, Carbon (0.03% vs. 18.5%), Hydrogen (0.14% vs. 9.5%), Oxygen (46.6% vs. 65.0%), Nitrogen (very little vs. 3.3%), and Sulphur (0.03% vs. 0.3%) are more abundant in the human body.

Practice Question 2

Single-Correct MCQ | Amino Acids

Identify the correct matching of the proteinaceous amino acid with its variable R-group or chemical classification:

(1) Alanine — Hydroxy methyl group ($-\mathrm{CH_2OH}$)

(2) Serine — Methyl group ($-\mathrm{CH_3}$)

(3) Valine — Neutral amino acid; Lysine — Basic amino acid; Glutamic acid — Acidic amino acid

(4) Tyrosine — Acidic amino acid; Tryptophan — Basic amino acid

Correct Answer: (3)

Worked Solution: Based on the number of amino and carboxyl groups, glutamic acid is an acidic amino acid, lysine is a basic amino acid, and valine is a neutral amino acid. Tyrosine, phenylalanine, and tryptophan are aromatic amino acids. In glycine $\mathrm{R} = -\mathrm{H}$, in alanine $\mathrm{R} = -\mathrm{CH_3}$ (methyl), and in serine $\mathrm{R} = -\mathrm{CH_2OH}$ (hydroxy methyl).

Practice Question 3

Multiple-Statement MCQ | Lipids

Read the following statements regarding lipids and select the option containing **only correct** statements:

I. Palmitic acid has 16 carbon atoms including the carboxyl carbon, whereas arachidonic acid has 20 carbon atoms including the carboxyl carbon.

II. Glycerol is chemically trihydroxy propane.

III. Oils such as gingelly oil have a higher melting point than fats and hence solidify in winters.

IV. Lecithin is a phosphorylated lipid (phospholipid) found in the cell membrane.

V. Lipids are polymeric macromolecules with molecular weights exceeding $10,000\ \mathrm{Da}$.

(1) I, II, and IV only

(2) I, II, III, and IV only

(3) II, IV, and V only

(4) I, IV, and V only

Correct Answer: (1)

Worked Solution: Statements I, II, and IV are completely accurate. Statement III is false because oils (like gingelly oil) have a *lower* melting point and remain liquid in winters. Statement V is false because lipids are *not* polymers and their molecular weights do not exceed $800\ \mathrm{Da}$.

Practice Question 4

Single-Correct MCQ | Nucleosides vs. Nucleotides

Which of the following groups contains **exclusively nucleosides**?

(1) Adenosine, Guanosine, Thymidylic acid, Uridine

(2) Adenine, Guanine, Cytosine, Thymine, Uracil

(3) Adenosine, Guanosine, Thymidine, Uridine, Cytidine

(4) Adenylic acid, Guanylic acid, Cytidylic acid, Uridylic acid

Correct Answer: (3)

Worked Solution: When nitrogenous bases are attached to a sugar, they are called nucleosides: Adenosine, Guanosine, Thymidine, Uridine, and Cytidine. When a phosphate group is also esterified to the sugar, they form nucleotides: Adenylic acid, Thymidylic acid, Guanylic acid, Uridylic acid, and Cytidylic acid.

Practice Question 5

Matching Type MCQ | Secondary Metabolites

Match Column I (Category of Secondary Metabolite) with Column II (Representative Examples) and choose the correct option:

Column I	Column II
A. Alkaloids	(i) Abrin, Ricin
B. Toxins	(ii) Concanavalin A
C. Lectins	(iii) Vinblastin, Curcumin
D. Drugs	(iv) Morphine, Codeine

(1) A-(iv), B-(i), C-(ii), D-(iii)

(2) A-(iv), B-(ii), C-(i), D-(iii)

(3) A-(iii), B-(i), C-(ii), D-(iv)

(4) A-(i), B-(iv), C-(iii), D-(ii)

Correct Answer: (1)

Worked Solution: According to the classification of secondary metabolites: Alkaloids = Morphine, Codeine (iv); Toxins = Abrin, Ricin (i); Lectins = Concanavalin A (ii); Drugs = Vinblastin, curcumin (iii).

Practice Question 6

Assertion–Reason | Acid-Insoluble Fraction

Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A): Although the molecular weight of lipids does not exceed $800\ \mathrm{Da}$, they get separated in the acid-insoluble macromolecular fraction during tissue grinding with trichloroacetic acid.

Reason (R): When a tissue is ground, cell membranes and other membranes are broken into pieces and form vesicles which are not water-soluble.

In the light of the above statements, choose the correct answer from the options given below:

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(3) (A) is true but (R) is false.

(4) (A) is false but (R) is true.

Correct Answer: (1)

Worked Solution: Lipids have small molecular weights ($\leq 800\ \mathrm{Da}$) and are not strictly macromolecules. However, because they are organized into membranes, grinding disrupts the membranes into water-insoluble vesicles that cannot pass through the filter into the acid-soluble pool, remaining instead in the acid-insoluble retentate. Thus, both (A) and (R) are true and (R) correctly explains (A).

Practice Question 7

Single-Correct MCQ | Cellular Composition

Arrange the following components of a living cell in **decreasing order** of their percentage of the total cellular mass:

A. Lipids | B. Carbohydrates | C. Nucleic acids | D. Proteins | E. Ions

(1) $\mathrm{D} > \mathrm{C} > \mathrm{B} > \mathrm{A} > \mathrm{E}$

(2) $\mathrm{D} > \mathrm{B} > \mathrm{C} > \mathrm{A} > \mathrm{E}$

(3) $\mathrm{C} > \mathrm{D} > \mathrm{B} > \mathrm{A} > \mathrm{E}$

(4) $\mathrm{D} > \mathrm{C} > \mathrm{A} > \mathrm{B} > \mathrm{E}$

Correct Answer: (1)

Worked Solution: The average composition of cells (% of total cellular mass) is: Water (70–90%) > Proteins ($\mathrm{D}: 10$–15%) > Nucleic acids ($\mathrm{C}: 5$–7%) > Carbohydrates ($\mathrm{B}: 3\%$) > Lipids ($\mathrm{A}: 2\%$) > Ions ($\mathrm{E}: 1\%$). Therefore, the correct decreasing order is $\mathrm{D} > \mathrm{C} > \mathrm{B} > \mathrm{A} > \mathrm{E}$.

Practice Question 8

Matching Type MCQ | Protein Functions

Match the proteins in Column I with their specific physiological functions in Column II:

Column I	Column II
A. Collagen	(i) Enables glucose transport into cells
B. Trypsin	(ii) Hormone
C. Insulin	(iii) Intercellular ground substance
D. GLUT-4	(iv) Enzyme

(1) A-(iii), B-(iv), C-(ii), D-(i)

(2) A-(iii), B-(ii), C-(iv), D-(i)

(3) A-(i), B-(iv), C-(ii), D-(iii)

(4) A-(iv), B-(iii), C-(i), D-(ii)

Correct Answer: (1)

Worked Solution: Collagen functions as intercellular ground substance (iii) and is the most abundant protein in the animal world; Trypsin is a proteolytic enzyme (iv); Insulin is a peptide hormone (ii); GLUT-4 enables glucose transport into cells (i).

Practice Question 9

Single-Correct MCQ | Polysaccharides

Which of the following statements regarding polysaccharides is **incorrect**?

(1) Starch forms helical secondary structures and holds I_2 molecules in the helical portion to give a blue colour.

(2) Cellulose is a homopolymer of glucose that lacks complex helices and therefore cannot hold I_2.

(3) In a glycogen chain, the left end is called the reducing end and the right end is called the non-reducing end.

(4) Inulin is a polymer of fructose, while chitin is a complex polysaccharide present in the exoskeletons of arthropods.

Correct Answer: (3)

Worked Solution: In a polysaccharide chain (such as glycogen), the **right end is called the reducing end** and the **left end is called the non-reducing end**. Statement (3) reverses these two ends and is therefore incorrect.

Practice Question 10

Single-Correct MCQ | Protein Structure

Consider the structural organization of proteins and select the **correct** statement:

(1) In a polypeptide chain, the first amino acid (left end) is the C-terminal amino acid and the last amino acid (right end) is the N-terminal amino acid.

(2) Both left-handed and right-handed helices are equally common in the secondary structure of proteins.

(3) The tertiary structure, folded like a hollow woolen ball, is absolutely necessary for the many biological activities of proteins.

(4) Adult human haemoglobin exhibits tertiary structure as its highest level of organization, consisting of three identical α-subunits.

Correct Answer: (3)

Worked Solution: Tertiary structure provides the 3-dimensional view (hollow woolen ball) and is absolutely necessary for the biological activities of proteins. Statement (1) is wrong because the left/first amino acid is N-terminal and the right/last is C-terminal. Statement (2) is wrong because in proteins *only right-handed helices* are observed. Statement (4) is wrong because adult human haemoglobin shows *quaternary* structure with 4 subunits ($2\alpha + 2\beta$).

Practice Question 11

Statement Evaluation | Enzyme Properties & Activation Energy

Given below are two statements:

Statement I: All enzymes are proteins without exception, and they get permanently denatured at temperatures above $40\ ^{\circ}\mathrm{C}$ in all living organisms including thermophiles.

Statement II: Whether a chemical reaction is exothermic (spontaneous) or endothermic (energy-requiring), the substrate S must pass through a higher-energy transition state, and enzymes accelerate the reaction by lowering the activation energy barrier.

In the light of the above statements, choose the correct answer:

(1) Both Statement I and Statement II are correct.

(2) Both Statement I and Statement II are incorrect.

(3) Statement I is correct but Statement II is incorrect.

(4) Statement I is incorrect but Statement II is correct.

Correct Answer: (4)

Worked Solution: Statement I is incorrect for two reasons: first, some nucleic acids behave like enzymes (called *ribozymes*, so only *almost* all enzymes are proteins); second, enzymes from thermophilic organisms (living in hot vents and sulphur springs) are thermally stable and retain their catalytic power even at high temperatures up to $80\ ^{\circ}\mathrm{C}$–$90\ ^{\circ}\mathrm{C}$. Statement II is completely correct.

Practice Question 12

Single-Correct MCQ | Enzyme Kinetics & Inhibition

Which of the following statements accurately describes the effect of temperature, substrate concentration, or inhibitors on enzyme activity?

(1) Low temperature permanently denatures the enzyme protein, whereas high temperature preserves it in a temporarily inactive state.

(2) Once $V_{\max}$ is reached, adding more substrate increases reaction velocity linearly because excess free enzyme molecules are always available.

(3) Inhibition of succinic dehydrogenase by malonate is an example of competitive inhibition because malonate closely resembles the substrate succinate in molecular structure.

(4) Competitive inhibitors bind irreversibly to the metal co-factor of carboxypeptidase and have no application against bacterial pathogens.

Correct Answer: (3)

Worked Solution: Malonate closely resembles the substrate succinate in structure and competes for the active site of succinic dehydrogenase—a classic example of competitive inhibition. Statement (1) is reversed: *low* temperature preserves the enzyme in a temporarily inactive state, whereas *high* temperature destroys activity by heat denaturation. Statement (2) is false because at $V_{\max}$ all enzyme molecules are saturated with substrate and no free enzyme molecules remain.

Practice Question 13

Matching Type MCQ | Classification of Enzymes

Enzymes are classified into 6 classes based on the type of reaction they catalyse. Match each Enzyme Class in Column I with the reaction it catalyses in Column II:

Column I (Enzyme Class)	Column II (Catalysed Reaction)
A. Transferases	(i) Removal of groups from substrates by mechanisms other than hydrolysis leaving double bonds ($C = C$)
B. Hydrolases	(ii) Linking together of two compounds by joining $C - O$, $C - S$, $C - N$, or $P - O$ bonds
C. Lyases	(iii) Transfer of a group G (other than hydrogen) between a pair of substrates ($S - G + S' \rightarrow S + S' - G$)
D. Ligases	(iv) Cleavage of ester, ether, peptide, glycosidic, $C - C$, C-halide, or $P - N$ bonds by addition of water

(1) A-(iii), B-(iv), C-(i), D-(ii)

(2) A-(iii), B-(i), C-(iv), D-(ii)

(3) A-(iv), B-(iii), C-(i), D-(ii)

(4) A-(ii), B-(iv), C-(i), D-(iii)

Correct Answer: (1)

Worked Solution: Transferases catalyse transfer of a group G other than hydrogen (iii); Hydrolases catalyse hydrolysis of ester, ether, peptide, glycosidic, $C - C$, C-halide, or $P - N$ bonds (iv); Lyases catalyse non-hydrolytic removal of groups leaving double bonds (i); Ligases catalyse linking together of 2 compounds (ii).

Practice Question 14

Single-Correct MCQ | Co-Factors

Identify the **correct** set of statements regarding apoenzyme and co-factors:

A. The protein portion of a cofactor-requiring enzyme is called the apoenzyme.

B. In peroxidase and catalase, which catalyse the breakdown of hydrogen peroxide to water and oxygen, haem is a tightly bound organic prosthetic group.

C. Co-enzymes are organic compounds whose association with the apoenzyme is only transient, and co-enzymes NAD and NADP contain the vitamin niacin.

D. Zinc is a metal ion co-factor for the proteolytic enzyme carboxypeptidase.

E. Removal of the co-factor from the enzyme has no effect on its catalytic activity.

(1) A, B, C, and D only

(2) A, B, and E only

(3) B, C, D, and E only

(4) A, C, and D only

> **Correct Answer: (1)**
>
> **Worked Solution:** Statements A, B, C, and D are all directly true. Statement E is false because *catalytic activity is completely lost when the co-factor is removed from the enzyme*, which proves that co-factors play a crucial role in catalytic activity.

7. Common Pitfalls & NEET UG Revision Checklist

High-Frequency Exam Pitfalls to Avoid

- **Trap 1 — Lipids as Macromolecules:** Never mark "Lipids have molecular weight $> 10,000\,\mathrm{Da}$" as true! Lipids have molecular weight $\leq 800\,\mathrm{Da}$ and are *not* polymers or true macromolecules; they only appear in the acid-insoluble fraction because disrupted membranes form water-insoluble vesicles.
- **Trap 2 — Homopolymer vs. Heteropolymer:** Proteins and nucleic acids are **heteropolymers** (built from 20 amino acids and 4 nucleotides, respectively). Cellulose, starch, glycogen, inulin, and chitin are **homopolymers**.
- **Trap 3 — Starch vs. Cellulose Iodine Test:** Starch gives a **blue colour with** I_2 because it forms helical secondary structures that trap I_2. Cellulose lacks complex helices and **does not** trap I_2.
- **Trap 4 — Glycogen Ends vs. Protein Ends:**
  - In **Polysaccharides (Glycogen)**: Left end = **Non-reducing end**; Right end = **Reducing end**.
  - In **Proteins**: Left end (first amino acid) = **N-terminal**; Right end (last amino acid) = **C-terminal**.
- **Trap 5 — Prosthetic Group vs. Co-enzyme:** Both are **organic compounds**! The distinction is binding affinity: **prosthetic groups** (e.g., haem in catalase/peroxidase) are *tightly bound*, whereas **co-enzymes** (e.g., NAD, NADP containing vitamin niacin) bind *transiently* during catalysis.

8. Concise Final Recap

- **Elemental Abundance:** $\mathrm{C}, \mathrm{H}, \mathrm{O}$ are relatively much higher in living tissues than in the Earth's crust. **Water** is the most abundant chemical in living cells (70–90%).

- **Acid-Soluble vs. Acid-Insoluble:** Grinding tissue in $\mathrm{Cl_3CCOOH}$ yields the acid-soluble filtrate (biomicromolecules, $18–800\,\mathrm{Da}$, plus inorganic ions like $\mathrm{SO_4^{2-}}, \mathrm{PO_4^{3-}}$) and the acid-insoluble retentate (proteins, nucleic acids, polysaccharides $\geq 10,000\,\mathrm{Da}$, plus membrane lipid vesicles).
- **Proteins:** Heteropolymers of 20 α-amino acids linked by peptide bonds. **Collagen** is the most abundant protein in the animal world; **RuBisCO** is the most abundant protein in the entire biosphere.
- **Polysaccharides & Nucleic Acids:** Cellulose, starch, and glycogen are homopolymers of glucose; inulin is a homopolymer of fructose; chitin is a complex homopolymer in arthropod exoskeletons. DNA (2'-deoxyribose) and RNA (ribose) are polynucleotides serving as genetic material.
- **Enzymes:** Mostly proteins (except catalytic RNA **ribozymes**) that lower activation energy via active sites in their tertiary structure. Classified into 6 classes (Oxidoreductases, Transferases, Hydrolases, Lyases, Isomerases, Ligases) and often require co-factors (prosthetic groups, co-enzymes, or metal ions) for catalytic activity.